Body Sensor Netowork Sessional

BME-422

IoT-Enabled IV Fluid Delivery System for Real-Time SPO2 Monitoring

Hafsa Khan
ID-1911004

Md Ebrahim
ID-1911006

Ummul Kainatt
ID-1911008

SUBMITTED TO
Ayesha Binth Humayun
Lecturer,
Department of Biomedical Engineering, CUET

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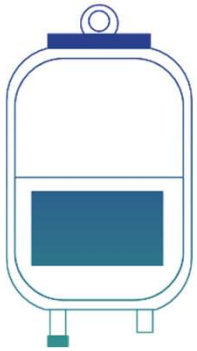
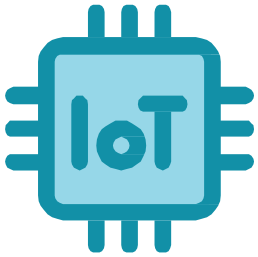
INTRODUCTION

- **SPO2 -** Percentage of oxygen in blood
- **Hypoxia** -Oxygen level deficiency in blood, which can primarily be cured using Acute **IV Fluid supply**
- **IoT (Internet of Things)** - communication between **devices and the cloud,** as well as between the devices themselves.





OBJECTIVE

- Introduce innovative **IoT-driven solution**
- Highlight R**eal-time SP02 monitoring**
- Emphasize **precision** in saline delivery
- Sustainable solution for **Hypoxia Patients**

Novelty OF THE PROJECT

Never Use Before

Low Production Cost

Use of IoT

Durability of Electronic Devices

LITERATURE REVIEW

Nurses' Role

Traditional System

Flow Adjustment

RESEARCH GAP

- **Precision** in **Saline Delivery**
- **Real-Time** Monitoring
- Remote **Monitoring**
- **Work Load and Error** Prevention

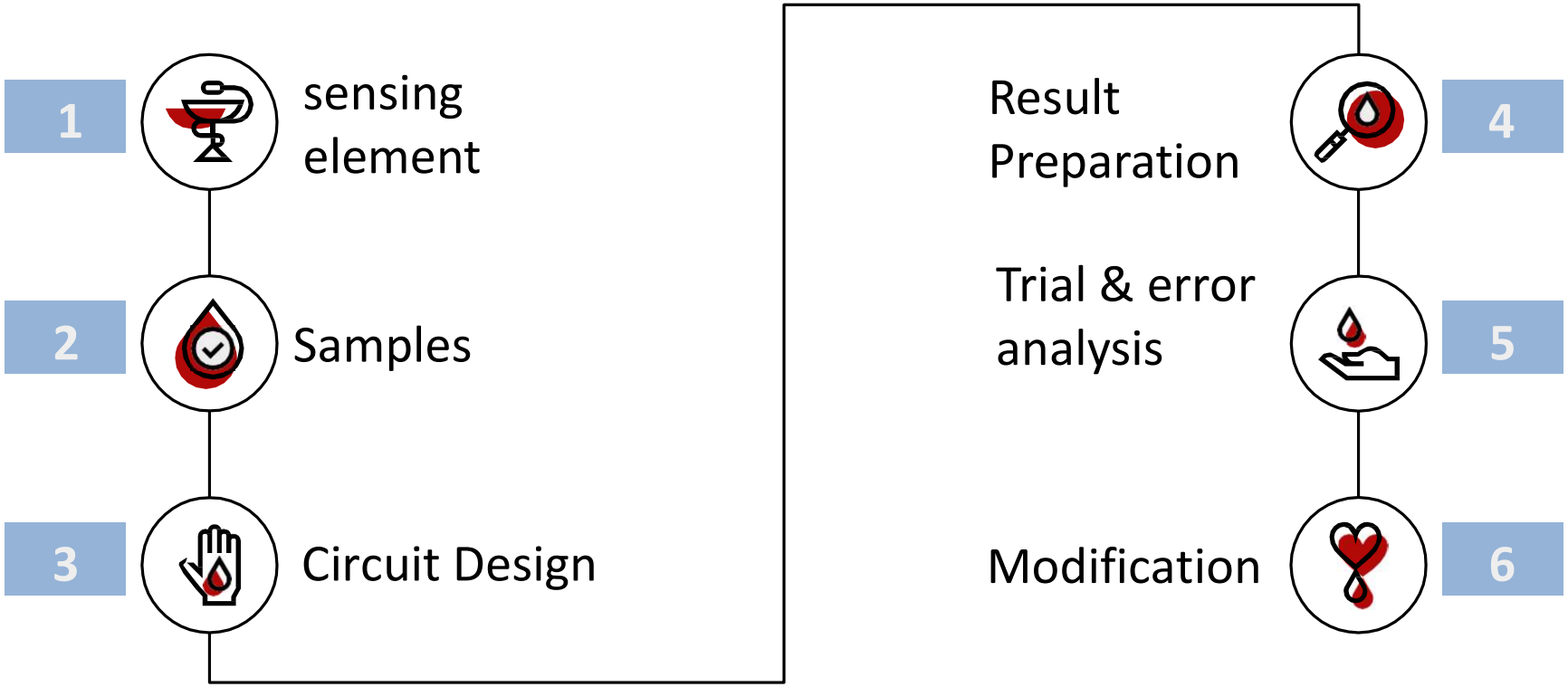
Prototype (Preferable solution)

Must Have Feature 1 → **To measure SpO2 by Sensing Element**

Must Have Feature 2 → **To estimate Fluid Quantity by flow measurement**

Additional feature 1 → **Creating a website for Database Record of Patient**

Materials and Methods



METHODOLOGY

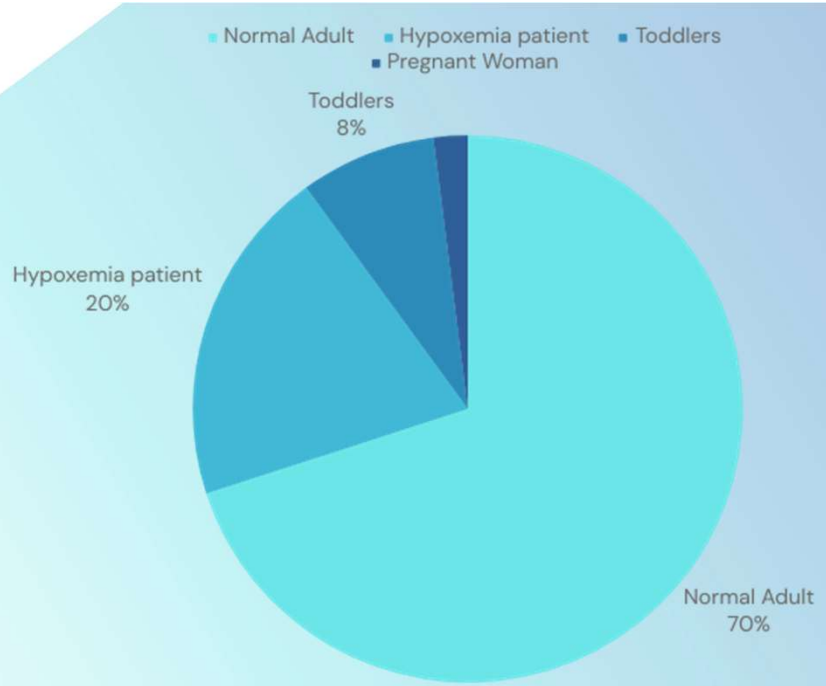
To solve the problem we followed some steps and used most efficient, feasible and reliable techniques

01 **Project Design:** Portable, Non-invasive , IoT based real time monitoring.

02 **Data sampling:** Mix sampling containing 70% normal adults, 20% Hypoxia Patients

03 **Variables:** Focuses on two key variables: **oxygen saturation level (SP02)** and necessary **saline volume**

04 **Data Analysis:** The experiment used **random selection, linear regression, ESP 8266** for data transfer to a web platform.



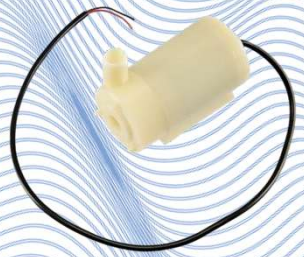
05 **Hypotheses:** A precision over **90%** with saline distribution safety.

06 **Pilot Study: The** system evolved from **Arduino to NodeMCU**, enhancing data accuracy and transfer.

KEY COMPONENTS

- **MAX30100** sensor
- **NodeMCU 8266** for IoT connectivity
- **Linear regression model** for adaptive saline delivery
- 5V **DC Pump with relay** switch to control saline flow
- **I2C Port** Connectivity

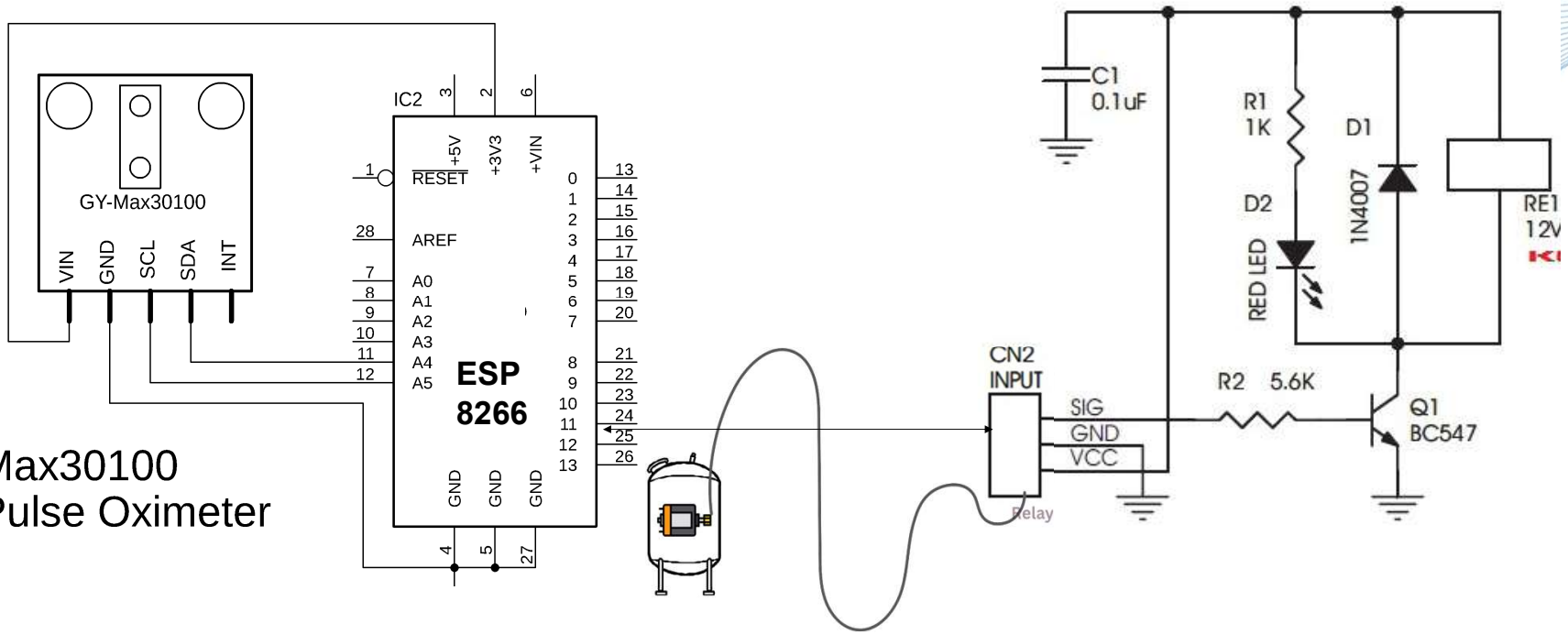
MAX30100



5V DC PUMP

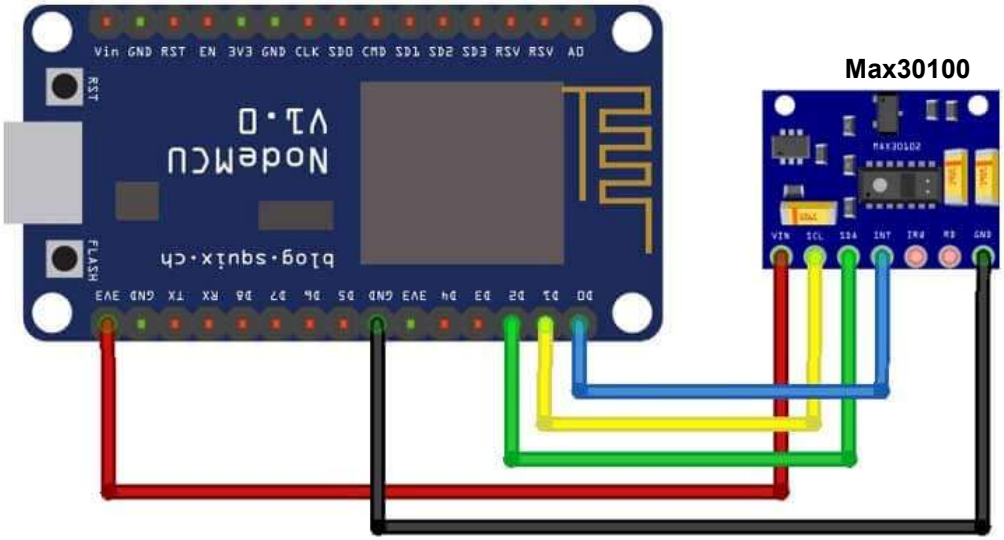
ESP8266

SCHEMATIC DIAGRAM



Max30100
Pulse Oximeter

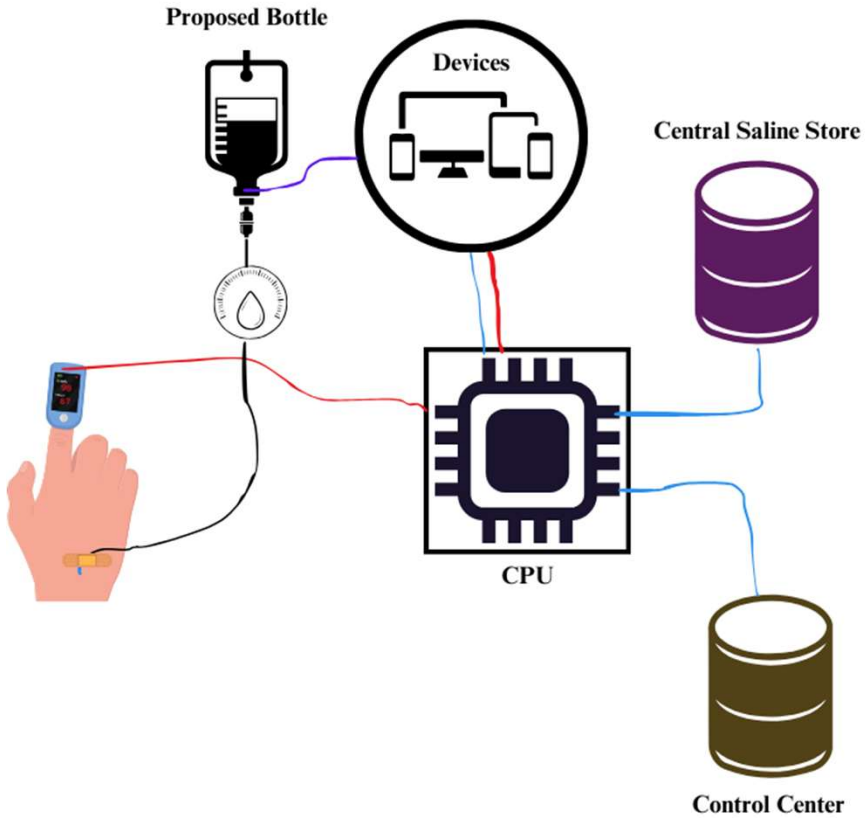
For Lab work


Max30100
NodeMCU
V1.0
blog.squix.ch
Vin GND RST EN 3V3 GND CLK SD0 CMD SD1 SD2 SD3 RSV RSV A0
RST
FLASH
3V3 GND TX RX D8 D7 D6 D5 D5 GND 3V3 D4 D3 D2 D1 D0
VIN SCL SDA INT IR4 R3 GND

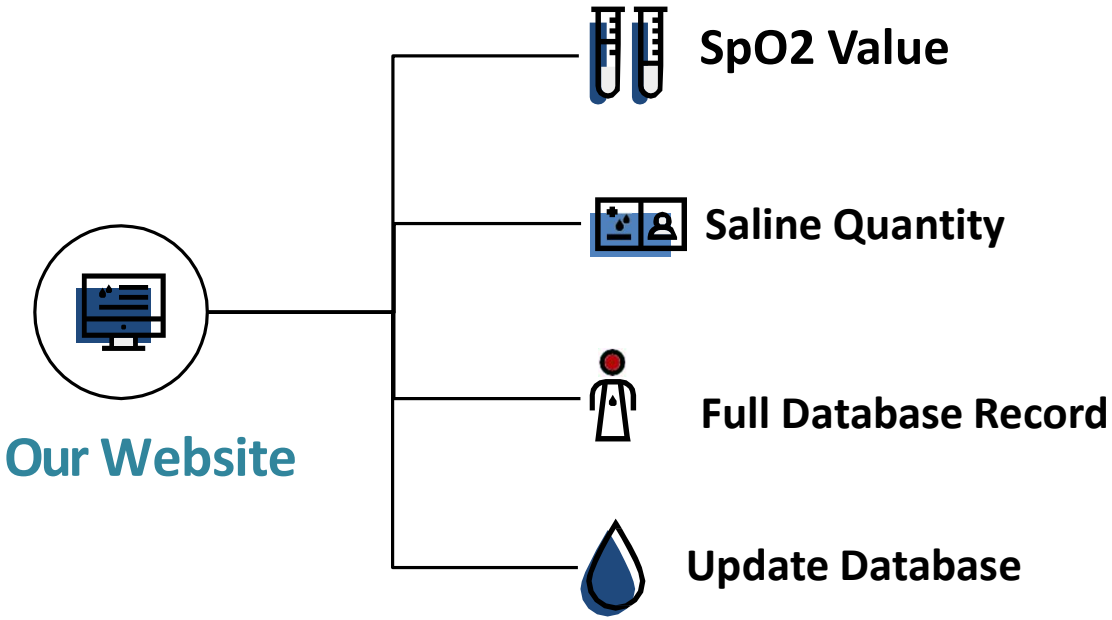
IoT Pulse Oxim
BPM
V1
SP02
V2
0
130
0
100
BPM
V1: --
SP02
V2: --

Future Work

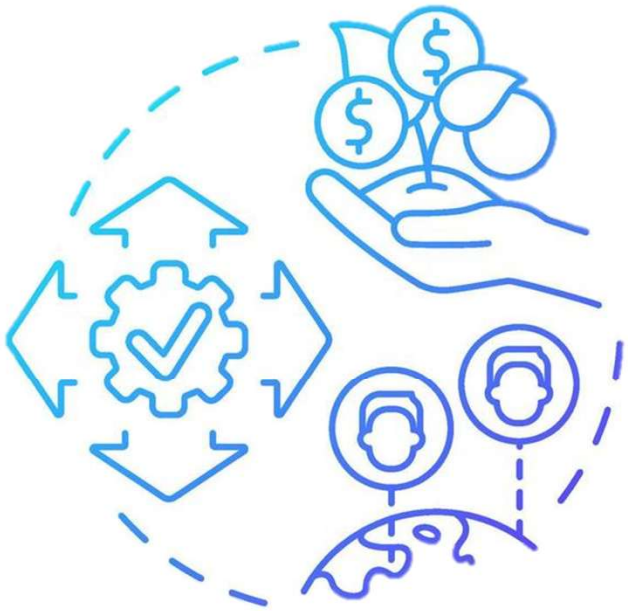
Circuit Diagram



Additional Features

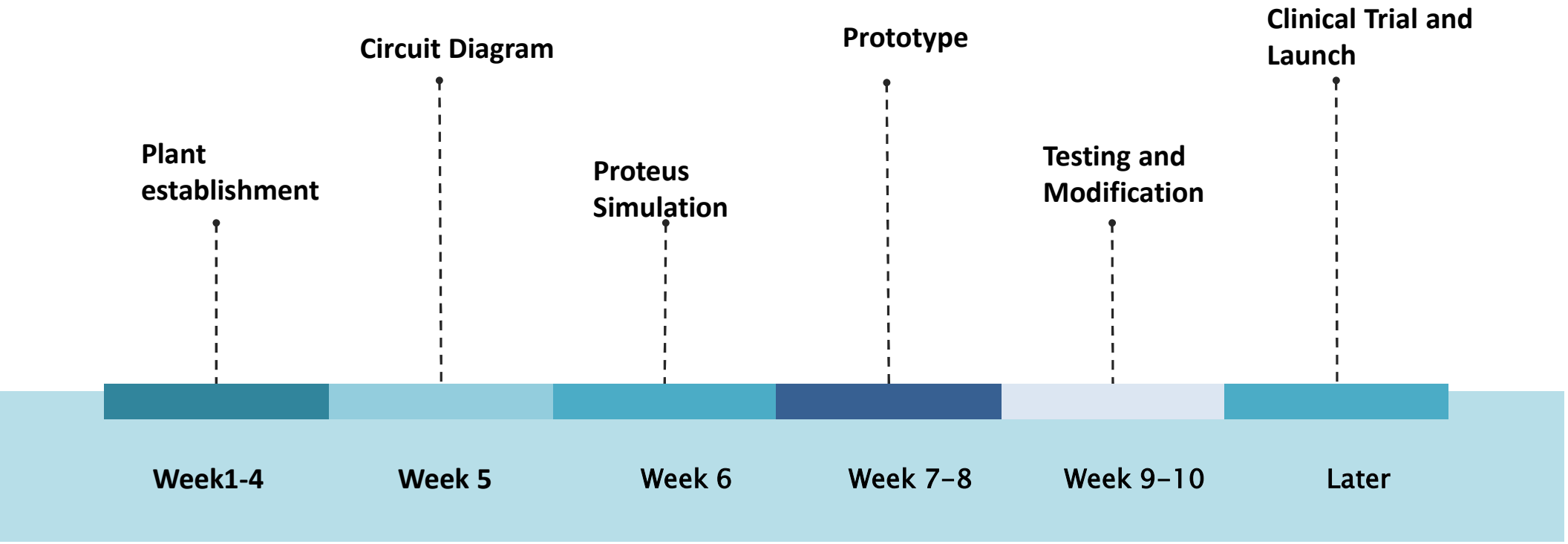


Commercial Potential



1. Hospitals & ICUs
2. Ambulance Services
3. Home Healthcare

PROJECT TIMELINE



FUTURE Scope

1. **Microlevel Saline Delivery**
2. **Smart Cleansing System**
3. **AI Integration**

CONCLUSION

- **IoT-driven Fluid delivery system**
- **precision in saline delivery**
- **Minimize Error**
- **Sustainable solution for Hypoxia Patients**

ACKNOWLEDGMENT

1 .W. OTTESTAD, J. I. KASIN, AND L. Ø. HØISETH, "ARTERIAL OXYGEN SATURATION, ° PULSE OXIMETRY, AND CEREBRAL AND TISSUE OXIMETRY IN HYPOBARIC HYPOXIA," AEROSPACE MEDICINE AND HUMAN PERFORMANCE, VOL. 89, NO. 12, PP. 1045–1049, 2018.

2 . R. REDDY, S. PRAKASH, A. DINAKAR, S. KUMAR, AND P. KODALI, "IOT BASED SALINE LEVEL MONITORING SYSTEM,"

3 . BAVISKAR, P. PATIL, S. BHATAMBRE, M. HAKE, AND S. ADSURE, "IOT BASED SALINE LEVEL MONITORING SYSTEM," TECHNOLOGY, VOL. 4, NO. 09, 2015.

4 .M. C. S. T. BHUVANESWARI, "SALINE LEVEL MONITORING SYSTEM," P. 1017–1025, SEP. 2022.

5. A. S. RAMTEKE, ""IOT BASED SMART SALINE LEVEL MONITORING SYSTEM"," MAY. 2021.

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